

Universidade de Aveiro
Doctoral Programme in Business and Economics
Advanced Corporate Finance (CFE)

Critical Review

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Table of Contents

Table of Contents	2
0. Justification for the Choice of Paper	5
0.1. Theoretical Relevance to Advanced Corporate Finance	5
0.2. Empirical and Methodological Innovation	5
0.3. Intersection with Value-Based Healthcare	5
0.4. Governance and Data Use	6
0.5. Contribution to Policy and Research Dialogue	6
Conclusion	6
Abstract	7
Executive Summary	8
Purpose and Context	8
Summary of the Article	8
Critical Evaluation	8
Theoretical Contribution	8
Methodological Advancement for the Empirical Continuation	9
Integration with Course Learning Outcomes	9
Expected Academic and Policy Contributions	9
Final Reflection	9
Structured, Standardized Keywords	11
Acronyms and Abbreviations	12
Presentation Slides	13
1. Introduction	14
1.1 Context and Relevance	14
1.2 Objective of the Review	14
1.3 Connection to Empirical Continuation	15
2. Summary of the Article	16
2.1 Research Question and Objectives	16
2.2 Data and Empirical Scope	16
2.3 Methodology Overview	17
2.4 Key Findings and Conclusions	18
3. Theoretical and Literature Foundations	21
3.1 Use of Corporate Finance Theory	21
3.2 Integration with Healthcare and Public Finance Research	22
3.3 Identification of Conceptual Gaps	23

4. Methodological Evaluation	25
4.1 Research Design Adequacy.....	25
4.2 Measurement and Operationalization	26
4.3 Data Quality and Reliability	27
4.4 Statistical Techniques and Analytical Soundness.....	28
4.5 Summary of Methodological Strengths and Weaknesses	30
5. Empirical Results and Interpretation.....	32
5.1 Consistency between Results and Claims	32
5.2 Alternative Explanations	33
5.3 Interpretive Limitations.....	34
5.4 Implications for Future Empirical Analysis	35
6. Contribution, Limitations, and Theoretical Value	37
6.1 Theoretical Contribution	37
6.2 Empirical Contribution and Generalizability.....	38
6.3 Identified Limitations.....	39
6.4 Relevance for Future Research	40
7. Recommendations for Improvement	43
7.1 Redefinition of Financial Distress	43
7.2 Methodological Enhancements	44
8. Integration with Advanced Corporate Finance and Empirical Continuation.....	49
8.1 Alignment with CFE Topics	49
8.2 Theoretical Link to Empirical Work	50
8.3 Methodological Transition.....	51
8.4 Expected Academic Contribution.....	52
9. Conclusion.....	54
9.1 Summary of Evaluation.....	54
9.2 Bridge to Empirical Study.....	54
9.3 Final Reflection	56
References	57
Annex A	59

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0. Justification for the Choice of Paper

The paper “*Rethinking financial distress in public healthcare: evidence from Greek hospitals’ governance attributes*” by **Stefanos Karakolias (2025)** offers a compelling bridge between **corporate finance theory** and the **governance and financial sustainability of healthcare systems**, making it highly relevant for review within the framework of **Advanced Corporate Finance** and directly aligned with the focus of my PhD on the **use of data in value-based healthcare (VBHC)**.

0.1. Theoretical Relevance to Advanced Corporate Finance

The study employs **Altman’s Z’’-score**, a cornerstone model in financial distress prediction, to assess the viability of public hospitals. This application of a **corporate finance model to public healthcare** exemplifies the transferability of financial decision-making tools across sectors - a core learning goal in the course. It challenges the traditional assumption that public entities cannot experience financial distress in the same way as private firms, highlighting the role of **financial structure, leverage, and liquidity** - all fundamental topics in corporate finance.

Moreover, the discussion on **governance attributes**, such as **manager gender, size, and specialty**, reflects how **agency theory and governance mechanisms** - key theoretical pillars of corporate finance - can influence financial performance and risk in healthcare institutions. This integration of **financial and governance dimensions** mirrors the analytical scope of corporate finance, where decision quality depends on both capital structure and managerial behavior.

0.2. Empirical and Methodological Innovation

The use of **real financial data from 90 public hospitals** and the adaptation of Altman’s model to a **non-profit, state-supported environment** underscores the methodological rigor and adaptability of quantitative finance approaches to complex social systems. This directly supports the course’s emphasis on **empirical financial modeling** and **data-driven decision-making**.

For a researcher focused on **data use in VBHC**, the paper provides an excellent case of how **financial indicators derived from administrative data** can inform broader performance and policy evaluations, despite the limitations of traditional financial models in non-market settings.

0.3. Intersection with Value-Based Healthcare

At its core, the paper questions the **meaning of financial health** in public hospitals, arguing that apparent financial viability may mask underlying **structural**

dependency on subsidies and delayed payments. This resonates with the **VBHC paradigm**, where value is redefined beyond short-term financial returns to encompass **sustainable outcomes, efficiency, and stakeholder equity.**

By identifying that traditional financial models fail to capture the “real” sustainability of health organizations, the paper implicitly calls for **new data frameworks** that integrate **financial, operational, and outcome-based metrics** - precisely the kind of multi-dimensional data integration central to my doctoral research.

0.4. Governance and Data Use

The study’s findings on **gender diversity and financial performance** align with the **corporate governance literature** discussed in the course, which emphasizes diversity as a driver of better risk management and sustainable decision-making.

For the PhD context, this invites reflection on how **governance and data transparency** shape not only financial outcomes but also **accountability and value creation** in healthcare - an essential aspect of building trust and efficiency in value-based systems.

0.5. Contribution to Policy and Research Dialogue

Finally, the paper’s **policy implications** - particularly the call to **redefine financial distress** in public healthcare and to disclose the **hidden financial dependencies** of hospitals - provide a strong conceptual foundation for **rethinking value and sustainability metrics** in health systems. These insights are directly transferable to ongoing research on **financial resilience and data-informed governance** in value-based health reforms across Europe.

Conclusion

In sum, Karakolias (2025) provides a **rich empirical and conceptual contribution** that connects **corporate finance theory, public sector governance, and healthcare management**, offering a relevant, data-driven, and critical lens for analyzing **financial sustainability in health systems.**

The paper thus serves as an ideal case study for both **Advanced Corporate Finance** and for expanding the conceptual foundations of my **PhD on data use in value-based healthcare**, illustrating how **financial modeling and governance analysis** can inform the pursuit of **sustainable, data-enabled healthcare value.**

Abstract

This critical review analyzes “*Rethinking Financial Distress in Public Healthcare: Evidence from Greek Hospitals’ Governance Attributes*” by **Stefanos Karakolias (2025)**, situating it within the theoretical and empirical foundations of **Advanced Corporate Finance (CFE)**. The paper offers a novel yet incomplete attempt to apply private-sector models of financial distress to public healthcare institutions operating under *soft budget constraints*. Using the **Altman Z"-score** across 90 Greek public hospitals, the study finds uniformly high solvency ratios (mean $Z = 15.7$) despite structural liquidity shortages, chronic arrears, and dependence on state subsidies.

This review concludes that *Karakolias (2025)* makes a **diagnostic contribution** rather than a **theoretical one**: it reveals the limitations of corporate finance models in non-market systems but fails to construct a new framework capable of explaining such anomalies. The paper’s **cross-sectional design**, **lack of causal identification**, and **simplistic governance operationalization** constrain its capacity to generalize or infer mechanisms. Nonetheless, its findings—particularly the paradox of “apparent solvency”—form a valuable empirical foundation for rethinking financial sustainability in public hospitals.

Building on this critique, the forthcoming **empirical continuation** extends the Greek analysis to **Portugal’s National Health Service (SNS)** using longitudinal data (2017–2024). It introduces the **Public Hospital Financial Sustainability Index (PHFSI)**, a multidimensional metric encompassing liquidity realization, subsidy dependence, stakeholder pressure, and performance quality. This dynamic model will employ **panel econometrics (fixed effects, dynamic GMM)** and **causal inference (difference-in-differences, instrumental variables)** to test whether governance quality—measured via a **Governance Quality Index (GQI)**—affects financial resilience and reduces dependence on state intervention.

Together, the critical review and its empirical continuation bridge **corporate finance theory and public health economics**, demonstrating how advanced financial methodologies can inform sustainability and governance reforms in mission-driven institutions. The work ultimately calls for a reconceptualization of financial distress as **fiscal vulnerability under institutional dependency**, expanding corporate finance beyond its market origins.

Executive Summary

Purpose and Context

This critical review forms part of the *Advanced Corporate Finance (CFE)* curriculum, designed to evaluate a scientific article through methodological critique and theoretical integration. The chosen paper-Karakolias (2025)-is particularly relevant because it merges financial distress theory with healthcare management, reflecting ongoing debates on public-sector sustainability, efficiency, and governance.

Summary of the Article

- **Objective:** To test whether Altman's Z"-score can measure financial distress in public hospitals and how governance attributes (size, location, specialty, gender) influence results.
- **Data:** Financial statements from 90 Greek public hospitals.
- **Methods:** Z"-score computation; Mann-Whitney and Kruskal-Wallis tests.
- **Findings:** All hospitals appear financially healthy, yet liquidity shortages and delayed receivables (average 1,485 days) indicate structural fragility.
- **Conclusion:** Classical financial distress models overstate solvency in public systems; financial distress must be "redefined."

Critical Evaluation

The review acknowledges the study's originality but identifies key shortcomings:

- **Theoretical gap:** No integration of *soft budget constraint theory* or agency models; the Z"-score is applied without conceptual adaptation.
- **Methodological limitation:** Cross-sectional design, descriptive tests, no causal identification, and lack of validation outcomes.
- **Governance simplification:** Binary or categorical variables fail to capture institutional complexity.
- **Interpretive weakness:** Correlations are presented as insights without distinguishing between managerial performance and accounting distortions.

Theoretical Contribution

The article's main theoretical relevance lies in revealing where **corporate finance theory ceases to function**-its models assume bankruptcy discipline, absent in public sectors. The recognition of this failure invites new definitions of distress rooted in dependency, liquidity management, and stakeholder transfer of risk. The review builds on this insight by proposing a **Stakeholder-Distributed Financial Distress Framework**, aligning with **Kornai's (1986)** soft budget constraint theory and contemporary sustainability paradigms.

Methodological Advancement for the Empirical Continuation

The empirical paper derived from this review addresses every methodological deficiency of the Greek study:

- **Dynamic modeling:** Uses **panel data (2017–2024)** to capture financial evolution over time.
- **New metrics:** Introduces the **Public Hospital Financial Sustainability Index (PHFSI)** as an alternative to bankruptcy-based scores.
- **Causal identification:** Employs econometric strategies such as **difference-in-differences (DiD)** and **instrumental variables (IV)** to estimate governance impacts.
- **Governance refinement:** Develops a **Governance Quality Index (GQI)** grounded in agency and stewardship theory.
- **Validation:** Tests predictive accuracy against observable fiscal interventions (bailouts, arrears clearance).

Integration with Course Learning Outcomes

The work directly aligns with the **CFE curriculum**:

- *Capital Structure Theory:* Challenges the concept of optimal leverage in tax-funded entities.
- *Financial Distress Modeling:* Extends classical bankruptcy models into fiscal dependency frameworks.
- *Governance and Agency:* Expands agency theory to multi-principal contexts.
- *Empirical Methods:* Demonstrates how econometric modeling validates theory in complex institutional systems.

Expected Academic and Policy Contributions

- **Conceptual:** Redefines financial distress as *sustainability imbalance* rather than default probability.
- **Methodological:** Offers a replicable empirical model (PHFSI) adaptable to different health systems.
- **Empirical:** Provides cross-national evidence comparing Greece and Portugal.
- **Policy:** Equips public managers and policymakers with analytical tools to detect early signs of fiscal stress and evaluate governance reforms.

Final Reflection

The review and its continuation collectively demonstrate that **corporate finance is not confined to markets-it is a discipline of resource optimization under uncertainty**, applicable wherever capital allocation and value creation occur.

Public hospitals, though shielded from bankruptcy, face financial pressures that require similar analytical rigor. Adapting financial models to these contexts enhances both their explanatory power and societal impact.

In closing, *Karakolias (2025)* is best understood as a **conceptual turning point**-a study that exposes the fragility of existing models while inspiring their reinvention. The ensuing empirical research responds to that challenge, transforming critique into innovation and extending corporate finance into the domain of **public value and institutional sustainability**.

Structured, Standardized Keywords

Agency Theory
Altman Z-Score
Corporate Governance
Critical Review
Empirical Finance
Financial Distress
Financial Sustainability
Greek Hospitals
Portuguese National Health Service (SNS)
Public Healthcare
Public Sector Finance
Soft Budget Constraint

Acronyms and Abbreviations

Acronym/Abbreviation	Full Term/Meaning
CFE	Advanced Corporate Finance
DEA	Data Envelopment Analysis
DiD	Difference-in-Differences
EOPYY	National Health Insurance Fund (Greece)
EP	Empirical Work
GMM	Generalized Method of Moments (Arellano-Bond)
GQI	Governance Quality Index
IPSAS	International Public Sector Accounting Standards
IV	Instrumental Variables
OECD	Organisation for Economic Co-operation and Development
PHFSI	Public Hospital Financial Sustainability Index
PREMs	Patient-Reported Experience Measures
PROMS	Patient-Reported Outcome Measures
ROC	Receiver Operating Characteristic
SBC	Soft Budget Constraint
SNS	Serviço Nacional de Saúde (Portugal's National Health Service)
TBD	To Be Determined
ULS	Unidades Locais de Saúde (Local Health Units, Portugal)
VBHC	Value-Based Healthcare

Presentation Slides

<https://20251024-cfe.vercel.app/>

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1. Introduction

1.1 Context and Relevance

Financial distress has long occupied a central place in **corporate finance theory**, shaping our understanding of firm value, capital structure, and governance. Classical frameworks-such as **Altman's Z-score model** (1968), the **trade-off theory** of capital structure, and **agency theory**-assume that firms operate in competitive markets, face bankruptcy risk, and seek profit maximization. Within such environments, financial ratios act as reliable indicators of distress because failure carries real consequences: loss of control, restructuring, or liquidation.

However, these assumptions collapse in **public and non-profit organizations**, particularly in **tax-funded healthcare systems** operating under *soft budget constraints* (Kornai, 1986). In these settings, the threat of bankruptcy is nonexistent, and chronic deficits are routinely absorbed by the state. Yet these organizations still face financial strain, resource scarcity, and governance challenges that affect efficiency and value creation. Understanding how to measure and manage such “non-terminal” distress therefore becomes a crucial question-one that lies at the intersection of **corporate finance** and **public management**.

This issue has gained renewed urgency in Europe's **Beveridgean health systems**, where post-pandemic pressures, aging populations, and fiscal tightening expose persistent hospital indebtedness. Governments continue to inject funds-often hundreds of millions of euros annually-to sustain operations. These recurring bailouts blur the boundary between *financial sustainability* and *financial dependence*. From a corporate finance perspective, this creates an opportunity to rethink traditional distress and capital structure theories under conditions of **guaranteed solvency but systemic inefficiency**.

It is within this theoretical and policy context that **Karakolias (2025)** contributes to the debate by applying Altman's Z"-score model to assess the financial health of Greek public hospitals. The paper claims that, despite apparently “healthy” financial ratios, these institutions remain reliant on government support, revealing the inadequacy of bankruptcy-based models in public sectors. This intersection between **financial modeling and healthcare governance** makes the article a pertinent case for critical examination in an **Advanced Corporate Finance** course, which emphasizes theory application, empirical rigor, and methodological soundness.

1.2 Objective of the Review

The purpose of this critical review is to evaluate **Karakolias (2025)** through the lens of corporate finance methodology and theory, assessing how convincingly the study adapts private-sector financial distress models to public healthcare organizations. The review will focus on three dimensions:

1. **Theoretical positioning** - whether the study effectively integrates corporate finance concepts (financial distress, capital structure, agency theory) with the institutional realities of public hospitals.

2. **Methodological robustness** - whether the chosen research design, variable operationalization, and statistical techniques support the author's conclusions.
3. **Contribution and limitations** - whether the article advances theoretical understanding or remains a descriptive application of existing tools.

Following the evaluation framework provided for this course, special emphasis is placed on the **methodological adequacy** of the empirical model-its capacity to produce valid, reliable, and generalizable results. The review will identify key strengths, expose conceptual inconsistencies, and suggest methodological improvements that align with contemporary corporate finance standards, such as **panel data modeling, causal inference, and multi-dimensional performance measurement**.

In doing so, this paper positions itself not as a mere critique but as a **constructive analytical bridge**: it seeks to transform the limitations of the reviewed study into research opportunities for deeper empirical inquiry.

1.3 Connection to Empirical Continuation

This critical review also serves as the **conceptual and methodological foundation** for the subsequent **Empirical Work (EP)** assignment of this course. The empirical paper will directly build upon the weaknesses identified in Karakolias (2025), extending the investigation from Greece to **Portugal's National Health Service (SNS)** using open financial and operational data from the national transparency portal (2017–2024).

Where Karakolias relied on a static, cross-sectional application of the Z"-score, the empirical study will develop a **Public Hospital Financial Sustainability Index (PHFSI)**-a dynamic indicator capturing liquidity, subsidy dependence, payment delays, and quality outcomes. The study will employ **panel regression, difference-in-differences, and instrumental variable** techniques to test causal relationships between governance structures and financial sustainability.

Thus, the review fulfills a dual role:

- It provides a **scholarly assessment** of an existing contribution at the intersection of finance and healthcare; and
- It lays the **analytical groundwork** for a new research project that adapts corporate finance theory to public systems with soft budget constraints.

By linking critical reflection to empirical innovation, this review embodies the pedagogical philosophy of **Advanced Corporate Finance**-to move from theory to critique, and from critique to evidence-based model building.

2. Summary of the Article

2.1 Research Question and Objectives

The article “*Rethinking Financial Distress in Public Healthcare: Evidence from Greek Hospitals’ Governance Attributes*” by **Stefanos Karakolias (2025)** explores the applicability of classical corporate finance models of financial distress—specifically **Altman’s Z"-score**—to the public healthcare sector. The author’s primary research question is straightforward yet provocative:

Can financial distress models originally designed for private, profit-oriented firms predict financial vulnerability in public hospitals that operate under state guarantees and soft budget constraints?

From this central question, the paper articulates several objectives:

1. **To assess the financial condition of Greek public hospitals** using the Z"-score model, thereby quantifying their apparent level of financial health.
2. **To examine whether governance characteristics**—including hospital size, geographic location, specialty, and CEO gender—correlate with differences in financial performance.
3. **To test the adequacy of traditional financial distress indicators** in capturing the economic reality of public healthcare institutions, and if necessary, to argue for a conceptual redefinition of “financial distress” in the public sector.

In essence, the study positions itself as a methodological experiment at the interface of corporate finance and public administration. It does not aim to create a new theoretical framework but to test the *transferability* of a canonical financial model from the corporate domain to public healthcare management. The author situates this inquiry within the broader debate on the financial sustainability of national health systems in post-crisis Southern Europe, where public hospitals face mounting liabilities, limited autonomy, and chronic dependence on state subsidies.

By confronting private-sector financial theory with the institutional realities of a tax-funded system, Karakolias seeks to illuminate how conventional corporate finance tools perform—or fail—when applied to organizations that cannot go bankrupt yet remain financially strained.

2.2 Data and Empirical Scope

The empirical analysis is based on a **cross-sectional dataset of 90 Greek public hospitals** operating under the Ministry of Health during the **fiscal year 2022**. This dataset represents the near-total population of large public healthcare providers

in Greece, encompassing both general and specialized hospitals across all administrative regions.

Financial data were obtained from officially audited statements submitted to the Ministry of Health's Financial Directorate. These include standardized balance sheets and income statements prepared according to national accounting standards, harmonized where possible with **International Public Sector Accounting Standards (IPSAS)**. The author reports adjusting specific items-such as depreciation, irregular income, and retained earnings-to improve comparability.

Governance data were extracted from administrative records and hospital websites, capturing categorical and binary variables:

- **Size** (small, medium, large) based on total assets and bed capacity,
- **Location** (urban vs. rural),
- **Specialty** (general, university-affiliated, or specialized), and
- **CEO gender** (male/female).

The sample year, 2022, was chosen because it provided the most recent complete financial statements available after the COVID-19 pandemic. However, the study deliberately adopts a **single-year cross-sectional design**, arguing that it allows a “snapshot” assessment of financial health and governance effects across institutions. No longitudinal or time-series components are included, and the analysis does not incorporate external control variables such as regional economic conditions, patient volume, or quality indicators.

This **narrow empirical scope** is both a strength and a weakness: it enables full-population coverage and comparability across hospitals but sacrifices the ability to capture temporal dynamics or causal relationships. The dataset thus serves primarily a **diagnostic purpose**, rather than one of forecasting or causal explanation.

2.3 Methodology Overview

The study employs a **quantitative analytical approach** centered on the **Altman Z"-score model**, originally developed to predict corporate bankruptcy in private manufacturing firms.

The Z"-score combines multiple financial ratios into a composite index intended to reflect a firm's likelihood of distress. In its standard form, the model is expressed as:

$$Z''=6.56X_1+3.26X_2+6.72X_3+1.05X_4$$

where:

- X_1 = Working Capital / Total Assets (liquidity),
- X_2 = Retained Earnings / Total Assets (profitability),
- X_3 = Earnings Before Interest and Taxes / Total Assets (operating efficiency),
- X_4 = Equity / Total Liabilities (solvency).

According to Altman's original thresholds, $Z'' < 1.1$ indicates high distress, $Z'' > 2.6$ indicates good financial health, and intermediate values define a "grey zone."

Karakolias applies this model directly to the financial statements of Greek public hospitals, without recalibration of coefficients, on the rationale that it provides a standardized and widely recognized measure of financial viability. The author acknowledges the conceptual tension of applying a private-sector bankruptcy model to public institutions but defends this decision as a diagnostic exercise intended to test whether such models are meaningful in non-market settings.

In addition to the Z'' -score calculation, the study conducts **bivariate statistical tests** to examine the relationship between financial health and governance attributes:

- **Mann-Whitney U tests** are used to compare median Z'' -scores between binary groups (e.g., male vs. female CEOs, urban vs. rural hospitals).
- **Kruskal-Wallis H tests** are applied for categorical variables with more than two levels (e.g., hospital specialty and size categories).

The analysis is complemented by **descriptive statistics** (means, standard deviations, coefficients of variation) for key financial indicators such as working capital, receivables, and overdue liabilities. No multivariate or regression-based models are estimated. The author argues that the non-parametric approach is suitable due to the small sample size and the non-normal distribution of financial ratios across institutions.

2.4 Key Findings and Conclusions

The empirical findings are striking and, in some respects, paradoxical.

All 90 hospitals exhibit **very high Z'' -scores**, with a **mean of 15.7** and a **minimum above 5.5**-far exceeding the "safe" threshold of 2.6. Based on the conventional Altman scale, every hospital in the sample would be classified as financially healthy, suggesting **zero probability of distress**. Yet, as the author points out, these institutions simultaneously display chronic liquidity shortages, extreme payment delays, and persistent reliance on state subsidies.

Key findings include:

1. **Financial viability is overstated** - High Z"-scores reflect accounting artifacts rather than true solvency, mainly because equity is inflated by government subsidies while current assets are dominated by uncollected receivables (93.5% of total).
2. **Governance patterns are statistically associated with Z"-score differences** -
 - Smaller hospitals and those located in rural areas tend to have higher Z"-scores.
 - Hospitals led by **female CEOs** also show higher scores, suggesting lower leverage or better liquidity management.
 - Hospital specialty (general vs. specialized) shows no significant difference.
3. **Working capital management is problematic** - Average receivable collection periods reach approximately **1,485 days**, implying that liquidity is largely illusory.
4. **Debt structure is atypical** - Virtually no long-term borrowing exists; liabilities are primarily short-term payables to suppliers.

From these findings, the author draws the overarching conclusion that **traditional financial distress models fail to capture the economic reality of public healthcare organizations**. Because such entities cannot go bankrupt, their high Z"-scores merely reflect **institutional dependency** on continuous state support. Karakolias therefore argues for a **“redefinition of financial distress”** in the public sector, proposing that future research should consider alternative indicators-such as liquidity risk, payment arrears, and subsidy reliance-as proxies for genuine fiscal vulnerability.

The paper concludes with a call for **new conceptual and methodological frameworks** capable of assessing the sustainability of public hospitals without relying on bankruptcy-based logic. It recommends future work that blends financial ratios with governance, operational, and contextual variables, emphasizing that financial performance in public healthcare must be interpreted through the lens of **mission-driven objectives and systemic interdependencies** rather than isolated accounting measures.

In summary, the study delivers an insightful diagnostic of the **limits of traditional corporate finance models** in the public sector and initiates a conversation on the need for **sector-specific approaches** to financial sustainability. However, as subsequent chapters of this review will argue, its theoretical and methodological execution leaves significant space for refinement-space that will be addressed through the **empirical continuation** using Portuguese data in the next stage of this coursework.

3. Theoretical and Literature Foundations

The strength of any empirical study in **corporate finance** lies not only in its analytical rigor but also in its capacity to position itself within established theoretical traditions. Karakolias (2025) aims to contribute to the ongoing dialogue between **financial theory and public healthcare management** by adapting the logic of corporate distress prediction to a context in which market forces and bankruptcy mechanisms are absent.

This chapter evaluates how effectively the paper draws from and integrates relevant theoretical frameworks—namely **financial distress theory**, **capital structure theory**, **agency theory**, and the **soft budget constraint paradigm**—and identifies key conceptual gaps that limit its theoretical contribution.

3.1 Use of Corporate Finance Theory

Karakolias (2025) anchors his analysis in one of the most enduring models of corporate failure prediction: **Altman’s Z-score framework** (Altman, 1968, 1993). This model, grounded in **discriminant analysis**, assumes that firms accumulate financial imbalances over time that can be detected through a composite of financial ratios—specifically those related to liquidity, profitability, leverage, and market value. Within corporate finance, such models have been used to quantify **default probability**, optimize **credit risk assessments**, and inform **capital structure decisions**.

However, the article **transplants this model directly** into the public hospital context without reinterpreting the theoretical assumptions that underpin it. In corporate finance, *financial distress* represents a transitional phase between solvency and bankruptcy, characterized by deteriorating cash flows, increased debt burden, and eroding market confidence. The mechanism culminates in **formal bankruptcy**, which acts as both a disciplinary and cleansing process within capital markets. In contrast, **public hospitals are insulated from bankruptcy**; their deficits are continuously refinanced by the state. This distinction challenges the basic premise that financial ratios can meaningfully predict “distress” when failure is institutionally precluded.

From the perspective of **capital structure theory**, the study implicitly touches on the **trade-off** and **pecking order** hypotheses but without explicit reference or theoretical adaptation. The observed near-zero levels of long-term debt and heavy reliance on subsidies reflect a **public-sector variant of the pecking order**, where internal and government-provided funds substitute for debt financing. Yet, the author does not connect this empirical observation to the broader literature on **optimal leverage** or to models that weigh tax shields against distress costs (Modigliani & Miller, 1958; Myers, 1984). Consequently, while the findings inadvertently support the notion of “zero optimal leverage” under conditions of tax exemption and guaranteed bailouts, this insight remains **unstated and under-theorized**.

Similarly, **agency theory**-central to corporate governance and financial performance-is only indirectly acknowledged. The paper measures governance attributes (gender, size, specialty) but does not theorize their relationship to financial outcomes through an agency or stewardship lens. In corporate finance, governance quality mitigates managerial opportunism and information asymmetry (Jensen & Meckling, 1976). In public hospitals, however, agency problems multiply due to **multiple principals** (government, regulators, patients) and the absence of residual claimants. By neglecting this theoretical complexity, the study reduces governance to descriptive variables rather than explanatory mechanisms.

Overall, the paper's use of corporate finance theory is **instrumental rather than integrative**-it employs the Z"-score as a computational tool but does not reinterpret its conceptual foundations to suit the institutional environment of public healthcare. The result is a technically competent application but a theoretically shallow engagement with corporate finance principles.

3.2 Integration with Healthcare and Public Finance Research

The paper acknowledges the specificities of the healthcare sector but engages only superficially with the **economic and institutional literature** that has long examined public hospital financing. The **soft budget constraint (SBC)** theory, developed by **János Kornai (1986)** and later refined by Maskin and Xu (2001), provides the most relevant conceptual lens for understanding fiscal behavior in state-owned enterprises and public institutions. It posits that when an organization anticipates rescue from its losses, the incentive for efficiency diminishes, leading to chronic overspending and dependency. In the context of public hospitals, this results in recurring arrears, supplier delays, and reliance on periodic government bailouts.

Karakolias (2025) briefly acknowledges that public hospitals operate with *soft constraints*, but this acknowledgment functions more as a disclaimer than as a theoretical foundation. The SBC framework is not operationalized or empirically tested. Instead of theorizing how subsidy dependence distorts financial ratios or undermines liquidity indicators, the paper simply observes these anomalies as empirical curiosities. Consequently, the model's inability to detect distress is interpreted as a statistical artifact rather than as a **predictable outcome of institutional incentives**.

The article also bypasses a rich body of **healthcare management and public finance research** that has explored financial sustainability in national health systems. Comparative studies from **Portugal (Matos et al., 2021)** and **Poland (Dubas-Jakóbczyk et al., 2020)** highlight structural issues-such as delayed payments, regulatory rigidity, and information asymmetry-that parallel the Greek case. These works propose multidimensional indicators incorporating operational, quality, and efficiency metrics, whereas Karakolias confines his analysis to accounting-based ratios.

Furthermore, there is limited engagement with the **public sector performance measurement** literature, which emphasizes outcome-based evaluation and value-for-money principles. Contemporary frameworks such as **value-based healthcare (Porter & Teisberg, 2006)** or **triple-bottom-line sustainability reporting (Elkington, 1997)** could have enriched the discussion by linking financial health to service quality and long-term value creation. The absence of such interdisciplinary integration narrows the paper's scope and confines it within the technical boundaries of ratio analysis, detached from the broader policy and managerial debates shaping public healthcare finance.

In short, while the study is situated in the healthcare context, it does not fully exploit the theoretical resources available in the **public finance** and **health management** literature. It treats the sector as an empirical setting rather than as an institutional system with unique economic logics that demand theoretical adaptation.

3.3 Identification of Conceptual Gaps

The combined evaluation of corporate finance and public sector integration reveals several critical **conceptual gaps** that constrain the explanatory and theoretical value of Karakolias's work. These gaps not only weaken the study's contribution but also define the research opportunities that the forthcoming empirical paper seeks to address.

1. Absence of a Reconciled Definition of Financial Distress

The article critiques the Z"-score's inability to capture true distress in public hospitals but stops short of proposing an alternative conceptualization. It fails to differentiate between **accounting solvency** (positive equity ratios) and **economic solvency** (sustainable cash flow). A more nuanced definition would consider "distress" as a multidimensional condition encompassing **liquidity risk, subsidy dependence, and stakeholder pressure**-dimensions that will be explicitly modeled in the empirical continuation through the **Public Hospital Financial Sustainability Index (PHFSI)**.

2. Lack of Dynamic Perspective

Financial distress is inherently a **temporal process**-firms transition from solvency to crisis over time. By relying on cross-sectional data, the study omits this dynamic evolution. The absence of a time component prevents the identification of early warning indicators or causal mechanisms linking governance to performance. The forthcoming empirical research addresses this by adopting a **panel data framework (2017–2024)** to analyze trajectories of distress and recovery.

3. Neglect of Governance Theory

The treatment of governance variables (size, gender, specialty) is descriptive and lacks theoretical anchoring. No conceptual link is drawn between governance structures and financial discipline, such as through **agency**, **stewardship**, or **institutional theories**. Future work will operationalize governance using a composite index reflecting **board financial expertise**, **managerial stability**, and **audit quality**, allowing for causal inference regarding their effects on sustainability.

4. Missing Link between Financial and Operational Performance

The study isolates financial ratios from operational or clinical outcomes. Yet, in public healthcare, financial efficiency is meaningful only insofar as it supports service delivery and patient outcomes. By ignoring this connection, the paper reduces the multidimensional nature of value creation to a single monetary dimension. The empirical paper will integrate **quality and access indicators** into PHFSI to bridge this divide.

5. Under-theorized Institutional Mechanisms

Finally, the article lacks a formal model explaining how soft budget constraints translate into measurable financial behaviors. Without specifying mechanisms—such as *expectations of bailout*, *transfer of distress to suppliers*, or *political interference in budgeting*—the paper remains descriptive. The forthcoming study proposes a “**stakeholder-distributed financial distress**” framework, wherein financial stress sequentially impacts suppliers, staff, patients, and taxpayers, forming a testable causal structure.

In conclusion, while Karakolias (2025) makes an important empirical observation—that conventional bankruptcy prediction models overstate financial health in public hospitals—it does not build the theoretical architecture necessary to explain *why* this occurs or *how* it could be remedied. Its theoretical engagement with corporate finance and public management is partial, and its integration across disciplines is limited.

These omissions are not failures but **starting points**: they justify the need for a new, data-driven framework that merges financial theory with the institutional realities of **Beveridgean healthcare systems**. This conceptual groundwork directly informs the design of the **empirical paper** that follows, which will operationalize these missing linkages through robust modeling and longitudinal evidence.

4. Methodological Evaluation

The methodological structure of Karakolias (2025) is central to understanding both the strengths and the limitations of its findings. The study presents itself as a quantitative assessment of public hospital financial health using the Altman Z"-score and non-parametric statistical testing. While it demonstrates procedural clarity and careful data compilation, its design choices constrain the interpretive power and theoretical depth of the analysis.

This chapter critically examines the research design, measurement validity, data reliability, and statistical methods employed, concluding with a synthesis that sets the stage for methodological advancement in the forthcoming empirical paper.

4.1 Research Design Adequacy

The study adopts a **cross-sectional design**, analyzing financial data from a single fiscal year (2022) for 90 Greek public hospitals. This design offers descriptive insights into the financial landscape of the Greek healthcare system but falls short of capturing the inherently **dynamic nature of financial distress**. In corporate finance, distress is conceptualized as a *process* of financial deterioration leading toward insolvency (Whitaker, 1999). Key mechanisms-such as declining profitability, growing leverage, and delayed payments-unfold over time. Thus, a static snapshot cannot reveal the temporal sequencing or causality underlying distress.

A cross-sectional design can be justified when the goal is benchmarking or classification. However, Karakolias (2025) positions his study as an exploration of **financial distress mechanisms**, a task requiring longitudinal data to identify early warning signals or predict intervention points. Without temporal variation, the model cannot distinguish between transient liquidity problems and structural insolvency, nor can it evaluate the persistence of financial vulnerabilities.

Moreover, the absence of a **control or comparison group** limits the study's explanatory scope. The paper could have enriched its design by comparing public and private hospitals, or by examining hospitals across different governance regimes. This would have allowed for quasi-experimental reasoning about how institutional ownership or autonomy affects financial outcomes.

Finally, the cross-sectional design restricts **causal inference**. Governance attributes (e.g., CEO gender or hospital size) may correlate with Z"-scores, but the direction of causality remains unknown-financially stronger hospitals might attract different types of leaders, rather than leadership traits determining financial outcomes. The absence of temporal or structural variation prevents testing for such reciprocal relationships.

In short, while the research design succeeds in providing a *comprehensive snapshot* of the sector's financial condition, it is not methodologically adequate

for addressing questions of *dynamics*, *causality*, or *mechanism*. These are precisely the dimensions that the subsequent empirical project-employing **panel data (2017–2024)** and **difference-in-differences (DiD)** methods-will aim to address.

4.2 Measurement and Operationalization

The core analytical instrument in the study is the **Altman Z"-score**, a composite index originally developed to predict bankruptcy risk in private-sector firms. The model integrates four financial ratios-liquidity (X_1), retained earnings (X_2), profitability (X_3), and solvency (X_4)-to estimate an organization's probability of default. The transfer of this model to public hospitals raises fundamental issues of **construct validity** and **contextual fit**.

Z"-score Adaptation

Karakolias applies the Z"-score *without re-specification of coefficients* or theoretical adjustment. This is problematic for several reasons:

1. **The dependent phenomenon (bankruptcy)** does not exist in public healthcare, where the state guarantees continuity of operations.
2. **Equity and retained earnings**, critical components of the Z" model, have distinct meanings in public accounting. For hospitals, "equity" includes accumulated subsidies and revaluations rather than shareholder capital, thereby inflating solvency ratios.
3. **Working capital** is distorted by receivables from state funds and health insurance entities, which often remain unpaid for years. Thus, liquidity ratios reflect accounting conventions rather than real cash availability.

Consequently, the Z"-score measures *accounting stability* rather than *financial sustainability*. A hospital can appear financially "healthy" while depending entirely on delayed subsidies and supplier credit. This mismatch between conceptual purpose and institutional context undermines the model's interpretive power.

Governance Variables

The operationalization of governance attributes is likewise limited. The study includes four categorical variables-size, location, specialty, and CEO gender-yet these proxies oversimplify complex managerial and institutional phenomena. For instance:

- “Size” is measured by total assets or bed capacity but ignores economies of scale, case-mix complexity, or service diversity.
- “Location” is treated as urban/rural, overlooking socio-economic disparities or regional funding differences.
- “Specialty” is a categorical label (general, university, specialized) with no adjustment for activity mix or teaching responsibilities.
- “CEO gender,” though statistically significant, is treated without theoretical grounding in agency, stewardship, or diversity-performance literature.

No measures of **board composition, financial expertise, audit mechanisms, or tenure stability**-core indicators in corporate governance research-are included. As a result, governance remains an *exogenous descriptor* rather than an *explanatory construct*.

Outcome Validity

Perhaps the most critical omission is the lack of a **validation outcome**. Because no hospital in the dataset declared bankruptcy, the model’s predictions cannot be tested against real distress events. The absence of a ground-truth benchmark means that the model’s accuracy and discriminatory power remain unverified. A robust empirical design would require alternative validation criteria-such as arrears accumulation, supplier defaults, or government bailouts-to test predictive validity.

4.3 Data Quality and Reliability

Karakolias (2025) draws data from **official audited financial statements** of Greek public hospitals, complemented by **administrative records** on governance attributes. This is a notable strength: the study benefits from full-population coverage and standardized accounting structures, ensuring internal consistency across units.

However, several challenges compromise data reliability and external validity:

1. Accounting Standards and Adjustments

The author claims to harmonize financial data with IPSAS principles but provides limited detail on how adjustments (e.g., depreciation, deferred revenue, extraordinary items) were made. Without transparent reporting of conversion rules, replication and comparability are impaired. Furthermore, IPSAS alignment does not resolve the fundamental difference between *cash flow timing* and *accrual recognition*-a critical issue when receivables remain unpaid for multiple years.

2. Receivables and Liquidity Illusion

The dataset includes inflated current assets due to outstanding payments from the national health insurance fund (EOPYY). The average receivable collection period of 1,485 days suggests that liquidity measures are unreliable proxies for real cash availability. This structural distortion inflates Z"-scores and biases the analysis toward false solvency.

3. Representativeness and Homogeneity

Although the dataset covers most hospitals, the institutions are highly heterogeneous-ranging from small regional units to large university teaching hospitals. The study applies the same financial model to all, ignoring differences in mission, case mix, and regulatory treatment. This lack of segmentation may mask systematic variation and reduce analytical precision.

4. Data Verification and Transparency

The study provides no appendix with descriptive statistics for individual variables or hospital-level outliers. Without this transparency, it is difficult to evaluate data accuracy, detect anomalies, or replicate findings. The author's reliance on aggregated summaries reduces methodological transparency-an essential requirement in modern empirical finance research.

In summary, while the data source is authoritative, its *interpretation* within a public-sector context introduces distortions that the paper does not adequately address. The forthcoming empirical study, by contrast, will leverage **open data from Portugal's SNS Transparency Portal** with explicit variable definitions, longitudinal consistency, and public reproducibility.

4.4 Statistical Techniques and Analytical Soundness

The analytical framework is intentionally simple: the author employs **non-parametric tests**-specifically the Mann-Whitney U test for two-group comparisons and the Kruskal-Wallis H test for multiple categories. This choice is justified by the non-normal distribution of financial ratios and the ordinal nature of some governance variables. While these tests are appropriate for exploratory analysis, they are limited in inferential capacity and fail to meet the methodological standards expected in advanced corporate finance research.

Key issues include:

1. Lack of Multivariate Analysis

The bivariate nature of the analysis cannot isolate the independent effect of each governance factor. For instance, hospital size and location may be correlated, leading to omitted variable bias. Standard practice in finance would employ **multiple regression, panel models, or structural equation modeling** to control for confounding variables.

2. No Causal Identification Strategy

Without longitudinal or exogenous variation, the study cannot distinguish causation from correlation. Modern empirical finance emphasizes **identification strategies**-such as natural experiments, instrumental variables, or difference-in-differences-to establish causal inference. The study's purely descriptive statistics offer associations but no evidence of mechanism.

3. No Sensitivity or Robustness Checks

The paper does not report robustness tests, alternative model specifications, or outlier diagnostics. This omission weakens confidence in the stability of results and raises the risk of spurious significance.

4. Overreliance on Arbitrary Thresholds

The interpretation of Z"-scores against Altman's original cutoffs (1.1, 2.6) assumes these thresholds remain valid across sectors and decades. This uncritical application ignores the need for recalibration to sector-specific distributions. Without empirical validation, threshold-based conclusions lack statistical justification.

5. Interpretive Circularity

The study's central claim-that high Z"-scores "mask" true distress-relies on subjective interpretation rather than statistical testing. The author infers failure from external observations (e.g., subsidy dependence) rather than from model performance metrics. This creates a circular argument: the model fails because the context differs, yet no quantitative evidence demonstrates this mismatch.

In essence, while the statistical analysis is transparent and straightforward, it remains **exploratory rather than explanatory**. Its simplicity underscores the feasibility of applying financial ratios to public data but limits its theoretical and predictive ambition.

4.5 Summary of Methodological Strengths and Weaknesses

Strengths

- Comprehensive coverage of Greek public hospitals, providing a representative sector overview.
- Clear, replicable computation of financial ratios.
- Transparent use of non-parametric methods suited for small, skewed samples.
- Identification of an important empirical paradox: high Z"-scores coexisting with structural insolvency.

Weaknesses

- Cross-sectional design unsuitable for studying dynamic distress processes.
- Conceptual misfit of the Z"-score in a non-bankrupt public sector.
- Simplistic operationalization of governance and lack of theoretical anchoring.
- No validation criterion for model performance.
- Absence of multivariate, causal, or robustness analysis.
- Limited transparency in data adjustment and reporting.

Implications for Replication and Extension

The methodological limitations observed here provide the **blueprint for improvement** in subsequent research. The forthcoming empirical paper will:

- Employ **panel data (2017–2024)** to capture temporal evolution and persistence of distress.
- Replace the Altman Z"-score with a **Public Hospital Financial Sustainability Index (PHFSI)** encompassing liquidity, subsidy dependence, payment delays, and quality indicators.
- Apply **multivariate fixed-effects and dynamic panel regressions** to control for confounders.
- Use **causal identification methods** (difference-in-differences and instrumental variables) to test governance effects.

- Ensure **data transparency and reproducibility** through open datasets and code publication.

Through these enhancements, the replication will not merely correct technical flaws but transform a static diagnostic study into a **dynamic, theory-driven model** of financial sustainability in Beveridgean healthcare systems.

5. Empirical Results and Interpretation

The empirical findings presented by Karakolias (2025) are both illuminating and paradoxical. They reveal the coexistence of apparently excellent financial ratios and structural fiscal fragility within Greek public hospitals—a contradiction that sits at the heart of the paper’s argument for “redefining financial distress.” However, while the results successfully demonstrate the inadequacy of the Altman Z"-score model in the public sector, they also raise questions about interpretation, causality, and completeness.

This chapter evaluates the internal coherence and credibility of these findings, considers alternative explanations, identifies interpretive limitations, and links them to future hypotheses for the empirical continuation using Portuguese data.

5.1 Consistency between Results and Claims

The study’s **main empirical finding**—that all 90 Greek public hospitals exhibit Z"-scores well above the threshold of financial distress (mean = 15.7, minimum > 5.5)—is clearly documented and statistically consistent. The descriptive statistics support the author’s central claim that, under conventional bankruptcy-based models, the entire sector appears financially sound. Yet this statistical outcome contradicts observable fiscal realities such as payment delays, subsidy dependence, and long-standing arrears.

This contradiction provides the foundation for the author’s conceptual argument: that **traditional distress models are not suitable for public entities**. In this sense, the data and conclusions are internally consistent—the results themselves exemplify the theoretical misfit between corporate finance models and public-sector behavior.

Nevertheless, the paper extends its interpretation beyond what the data can substantiate. The claim that **financial distress must be “redefined”** is logically valid but empirically untested within the study. The analysis identifies *symptoms* (e.g., inflated liquidity, zero long-term debt) but does not demonstrate *mechanisms* (e.g., subsidy dependence dynamics or fiscal softening effects). The results confirm a measurement failure, but not the proposed theoretical remedy.

Similarly, the observed **governance effects**—that smaller and female-led hospitals report higher Z"-scores—are reported as statistically significant but weakly interpreted. The paper infers that these institutions may be managed more prudently or operate with less leverage. However, given the earlier critique of the Z"-score’s contextual validity, any conclusion about managerial quality based on this indicator is fragile. It is logically inconsistent to argue both that the model is invalid and that its outcomes reflect genuine governance differences. Without alternative metrics or robustness tests, the link between governance and financial health remains speculative.

In sum, while the results are internally coherent as an empirical demonstration of the Z"-score's limitations, they are not theoretically or causally persuasive. The author's conclusions exceed what the data can meaningfully support, turning statistical anomalies into normative claims without additional evidence.

5.2 Alternative Explanations

The paper attributes observed variations in Z"-scores primarily to governance characteristics and institutional inefficiencies, but it omits several plausible **alternative explanations** that could account for the results:

1. Accounting and Structural Factors

The Z"-scores may be inflated not by managerial performance but by **accounting conventions** in the public sector. For instance, hospitals classify state subsidies and capital transfers as equity, thereby increasing solvency ratios. Moreover, delayed receivables from the national insurance fund (EOPYY) inflate working capital, giving the illusion of liquidity. These accounting treatments, rather than operational performance, likely drive the uniformly high Z"-scores.

2. Regulatory Homogeneity

All hospitals operate under the same Ministry of Health budgetary framework, with standardized funding mechanisms, procurement rules, and price ceilings. Such institutional uniformity limits governance variability and may homogenize financial outcomes. Observed score differences could therefore reflect minor accounting idiosyncrasies rather than substantive management effects.

3. Macro-Fiscal Conditions

The study does not account for macroeconomic or fiscal variables-such as central government arrears clearance programs or fiscal transfers from the European Union-which significantly influence hospital liquidity. Temporary debt forgiveness or late-year subsidies can artificially boost financial ratios. Without controlling for these exogenous shocks, it is impossible to determine whether high Z"-scores reflect systemic fiscal interventions or intrinsic hospital performance.

4. Operational Complexity and Case Mix

Hospitals differ in patient volumes, case-mix complexity, and teaching status. Larger or university hospitals typically bear higher fixed costs and longer revenue cycles due to complex billing and reimbursement processes. Their lower Z"-scores could therefore reflect operational intensity rather than inefficiency. The omission of operational variables skews interpretations of size and specialty effects.

5. Endogeneity of Governance Attributes

Governance variables, such as CEO gender or hospital size, may themselves be endogenous. For instance, female CEOs might be more likely to be appointed to smaller or less complex hospitals, which naturally present lower financial risks. Without addressing this **selection bias**, the observed relationship between gender and Z"-scores risks being spurious.

These omitted factors collectively suggest that the results, while numerically sound, are **conceptually underdetermined**. The study interprets correlation as explanation without accounting for structural, contextual, or endogeneity effects-limitations that will be explicitly addressed in the subsequent empirical analysis.

5.3 Interpretive Limitations

The interpretive weaknesses of the paper stem from a combination of **model misfit, overgeneralization, and causal ambiguity**.

First, the study falls into what can be termed *interpretive circularity*. It uses the Z"-score to demonstrate that the model itself is inappropriate, yet simultaneously interprets its outputs to infer patterns of governance and performance. This dual use of a model both as an object of critique and as an explanatory tool undermines the credibility of derived conclusions. A more rigorous approach would have entailed applying multiple models or testing sensitivity to ratio weighting.

Second, the absence of a validation outcome renders the analysis non-falsifiable. The author claims that high Z"-scores “mask true distress,” but this assertion cannot be tested without independent indicators-such as arrears growth, supplier insolvencies, or government bailouts. Without such benchmarks, the study substitutes interpretive reasoning for empirical proof.

Third, reverse causality poses a substantial challenge. For example, smaller hospitals may exhibit higher Z"-scores not because they are better managed but because they have lower leverage and simpler operations. Similarly, gender differences in financial outcomes may reflect structural biases in appointment processes rather than managerial behavior. Without a temporal framework or instrumental variables, the study cannot distinguish cause from correlation.

Fourth, overgeneralization limits external validity. The author extrapolates findings from Greece to all Beveridgean systems, yet contextual factors-such as Portugal’s stronger financial governance, the UK’s NHS structure, or Scandinavia’s decentralized funding-fundamentally alter financial behavior. Without cross-country comparison or theoretical boundary conditions, generalization remains speculative.

Finally, the interpretive framework is **monodimensional**, focusing exclusively on financial ratios. Public hospital performance is multidimensional, encompassing efficiency, quality, and equity. The claim that financial distress should be “redefined” implicitly acknowledges this multidimensionality, yet the study does not integrate any non-financial indicators to support the argument. Consequently, the interpretation lacks the holistic perspective necessary for understanding financial sustainability in mission-driven institutions.

5.4 Implications for Future Empirical Analysis

The inconsistencies and interpretive gaps identified in Karakolias (2025) directly inform the design of the forthcoming **empirical continuation** within this course. The new research, to be conducted using Portugal’s **SNS Transparency Portal** data (2017–2024), will address the shortcomings of the Greek study through explicit methodological and theoretical enhancements.

1. Dynamic Modeling of Financial Sustainability

Rather than relying on a single-year cross-section, the empirical paper will construct a **panel dataset** covering multiple years, enabling the observation of trends and persistence in financial distress. This will allow for **temporal causality** testing and early warning modeling-features absent in the Greek study.

2. Development of the Public Hospital Financial Sustainability Index (PHFSI)

The PHFSI will replace the Altman Z"-score as a **context-appropriate composite indicator**. It will incorporate measures of operational self-sufficiency, liquidity realization, subsidy dependence, and stakeholder pressure (supplier arrears, complaints, and quality outcomes). This multidimensional approach will operationalize the very “redefinition” of financial distress that Karakolias calls for but does not implement.

3. Causal Inference and Identification Strategies

Using longitudinal data, the Portuguese study will employ **fixed-effects regressions, dynamic panels (Arellano–Bond GMM), and difference-in-differences (DiD)** techniques. These methods will isolate the impact of governance reforms-such as the 2024 integration of hospitals into *Unidades Locais de Saúde (ULS)*-on financial sustainability. This addresses both reverse causality and omitted variable bias.

4. Governance Measurement Enhancement

Governance quality will be captured through composite indices measuring board expertise, managerial stability, audit effectiveness, and gender diversity. This replaces simplistic categorical variables with theoretically grounded constructs aligned with **agency and stewardship theories**.

5. Validation through Real Outcomes

The empirical work will test whether the PHFSI predicts **government interventions**, such as capital injections or bailouts-providing an objective validation criterion that the Greek study lacked. Predictive accuracy will be evaluated using **Receiver Operating Characteristic (ROC)** analysis, ensuring empirical falsifiability.

6. Cross-System Learning

By applying the new model to a different Beveridgean system (Portugal) and comparing results with those from Greece, the study will assess whether observed anomalies are **universal institutional features** or **country-specific artifacts**. This comparative perspective will bring theoretical depth and international relevance to the debate initiated by Karakolias.

In short, the Greek paper's results serve as both a warning and an invitation. They warn of the perils of transferring private-sector models uncritically to public contexts, yet they invite a new generation of research that combines **financial theory, governance analysis, and empirical rigor**. The forthcoming empirical paper will take up this invitation, transforming descriptive contradictions into testable hypotheses and advancing a **dynamic theory of financial sustainability in public healthcare**.

6. Contribution, Limitations, and Theoretical Value

Karakolias (2025) situates itself at a meaningful intersection between **corporate finance** and **public healthcare economics**, addressing an underexplored question: can private-sector financial distress models capture the fiscal realities of public hospitals? The study's ambition to bridge these two domains is commendable and timely. However, its theoretical reach and methodological design limit the extent to which it advances knowledge in either field.

This chapter evaluates the paper's contributions to theory and empirical understanding, outlines its limitations, and highlights how these weaknesses create the intellectual foundation for a more rigorous, data-driven framework-one that the forthcoming empirical work intends to build.

6.1 Theoretical Contribution

The paper's theoretical ambition lies in its **attempt to reinterpret corporate finance concepts**-particularly financial distress and governance-within a public-sector setting. By applying the **Altman Z"-score** to Greek hospitals, Karakolias brings attention to the conceptual misalignment between *market-driven insolvency models* and *state-guaranteed entities*. This recognition alone constitutes a modest theoretical contribution: it exposes a fundamental boundary condition in corporate finance theory.

However, the study stops short of **formulating new theory** or modifying existing frameworks to account for the distinct features of public institutions. It identifies an anomaly (public hospitals appearing solvent despite chronic deficits) but does not theorize the mechanisms that generate it. As a result, the paper **applies** corporate finance models rather than **extends** them.

A deeper theoretical engagement could have been achieved through three avenues that remain underdeveloped:

1. **Soft Budget Constraint Integration** – The paper alludes to public hospitals' reliance on state funding but fails to embed this within Kornai's (1986) theory of *soft budget constraints (SBC)*. In corporate finance terms, SBC represents the erosion of the disciplinary function of debt and default. Integrating SBC into the Z"-score framework could have produced a new conceptual hybrid-one where distress is defined not by bankruptcy risk but by *dependency risk* (the expectation of external rescue).
2. **Capital Structure Theory Adaptation** – The study implicitly supports the notion that optimal leverage in public hospitals is near zero, since debt costs outweigh nonexistent tax benefits. Yet this insight is not articulated in theoretical form. A modified **trade-off theory** could explain why state-

owned entities maintain high “equity ratios” that actually conceal subsidy accumulation rather than retained earnings.

3. **Governance and Agency Mechanisms** – The inclusion of gender, size, and specialty as governance proxies suggests an interest in **agency theory**, but the theoretical mechanism linking these variables to financial outcomes is missing. The public sector introduces unique agency problems-multiple principals (government, taxpayers, patients) and weak performance incentives-that could have redefined traditional governance-finance linkages.

In short, Karakolias (2025) contributes to **the critique of theoretical applicability** rather than to **the construction of new theory**. It performs a diagnostic function-revealing where classical corporate finance models fail-but leaves the theoretical reconstruction to future research.

The forthcoming empirical paper responds directly to this gap by proposing the **Stakeholder-Distributed Financial Distress Framework** and the **Public Hospital Financial Sustainability Index (PHFSI)**, both designed to extend corporate finance logic into non-market institutional environments.

6.2 Empirical Contribution and Generalizability

Empirically, the paper’s primary contribution lies in its **comprehensive quantitative mapping of Greek public hospital finances**. Using official data for 90 institutions, it provides one of the few full-population analyses of a Beveridgean system’s financial condition post-COVID-19. The study’s descriptive results-especially the high Z"-scores (mean 15.7), the extreme receivable delays (1,485 days), and the absence of long-term debt-offer valuable documentation of systemic fiscal patterns that policymakers and researchers often discuss only anecdotally.

These findings serve two empirical functions:

1. **Benchmarking:** They establish baseline indicators of the Greek health system’s financial condition in the early 2020s, which can support comparative research in other countries.
2. **Anomaly Identification:** They empirically demonstrate the disjunction between accounting solvency and operational fragility, paving the way for the concept of “apparent solvency” or “masked distress.”

However, the study’s **empirical generalizability** is limited by three key factors:

1. **Institutional Specificity:** Greece's fiscal and governance environment is unique, shaped by the legacy of the sovereign debt crisis, EU oversight, and centralized budget control. Results cannot be directly extrapolated to countries with different governance models (e.g., Portugal, the UK, or Scandinavia) without accounting for contextual heterogeneity.
2. **Lack of Comparative Analysis:** Without a benchmark population (such as private hospitals or hospitals from other health systems), the study cannot isolate which results are specific to Greek institutions versus universal features of public healthcare finance.
3. **Static Design:** The single-year dataset prevents analysis of trends, persistence, or responsiveness to policy reforms. This restricts the study's ability to generalize mechanisms of financial distress over time or across varying fiscal regimes.

Despite these limitations, the paper's empirical contribution should not be dismissed. It performs a **data-driven critique** that inspires further research—demonstrating both the promise and the pitfalls of applying corporate financial metrics to public systems. It provides a valuable empirical reference point for cross-national replication and theoretical innovation.

6.3 Identified Limitations

The paper's limitations span theoretical, methodological, and interpretive domains, reinforcing its characterization as a *diagnostic* rather than *explanatory* study. The most significant weaknesses include:

1. Conceptual Misfit of the Z"-score Model

The Z"-score assumes the existence of bankruptcy as an endpoint of financial deterioration. In a state-owned healthcare system, distress manifests not as bankruptcy but as arrears accumulation, subsidy dependence, or service degradation. The model's underlying logic—predicting default risk—is therefore incompatible with the institutional context.

2. Static, Cross-Sectional Design

Financial distress is inherently dynamic, evolving through time as organizations adjust to fiscal shocks and policy reforms. By relying on one year of data, the study captures a snapshot of ratios but misses the temporal sequence that defines

distress. This design also precludes testing causal relationships between governance and financial performance.

3. Weak Operationalization of Governance

The use of categorical proxies (gender, size, location, specialty) offers little insight into managerial quality, board competence, or incentive structures. Without theoretically grounded governance metrics (e.g., audit frequency, board independence, tenure), the paper cannot evaluate the mechanisms through which governance affects financial outcomes.

4. Absence of Validation Criteria

The author claims that the Z"-score fails to represent "true distress," yet provides no independent criterion—such as arrears growth, delayed supplier payments, or government interventions—to substantiate this assertion. The lack of validation leaves conclusions interpretive rather than empirical.

5. Overgeneralization and Lack of Comparative Context

The findings are extrapolated to all Beveridgean systems, but without controlling for institutional variation. Each national health service has distinct financing mechanisms, making unqualified generalization problematic.

6. Omission of Non-Financial Performance Dimensions

Financial health in healthcare organizations cannot be detached from service quality, access, and efficiency. By focusing exclusively on accounting indicators, the study neglects the multidimensional nature of sustainability, a limitation that reduces its relevance to policymakers and management scholars alike.

Collectively, these limitations underscore the need for a **new theoretical and empirical architecture**—one capable of integrating financial, operational, and governance data within a dynamic and causally robust framework.

6.4 Relevance for Future Research

The weaknesses outlined above do not diminish the importance of Karakolias's study; rather, they define the **agenda for the next generation of research** on public-sector financial sustainability. They point directly to the rationale for the **empirical continuation** to be conducted as part of this course. Four lines of relevance stand out:

1. Redefining Financial Distress through Multi-Dimensional Metrics

The forthcoming empirical paper will operationalize financial sustainability not through bankruptcy-based ratios but through a **Public Hospital Financial Sustainability Index (PHFSI)** that captures liquidity realization, subsidy dependence, stakeholder pressure, and clinical performance. This model transforms the conceptual critique raised by Karakolias into measurable constructs.

2. Dynamic and Causal Analysis

To overcome the static limitations of the Greek study, the new research will employ **panel data (2017–2024)** from Portuguese hospitals, allowing for the analysis of trends and persistence. Advanced econometric techniques—**fixed effects, dynamic GMM, and difference-in-differences (DiD)**—will enable causal inference regarding governance reforms and policy interventions (e.g., 2024 ULS integration reform).

3. Governance Theory Integration

Building on agency and stewardship theories, governance will be conceptualized as a **multifactor construct** involving financial expertise, audit mechanisms, managerial stability, and gender diversity. This theoretically grounded operationalization addresses the ad hoc treatment of governance in the Greek paper.

4. Validation through Real Fiscal Outcomes

The empirical continuation will link financial metrics to **observable intervention events**—capital injections, bailouts, or restructuring orders—thus providing objective validation and predictive testing absent from the Greek study.

5. Comparative Institutional Learning

By contrasting Portugal and Greece, the new research will assess whether the observed “apparent solvency paradox” is systemic to Beveridgean models or context-specific to fiscal governance structures. This comparative approach will contribute to theory-building on financial resilience in public systems.

Ultimately, Karakolias (2025) serves as both a **cautionary tale and a catalyst**. It cautions against the uncritical transposition of private-sector finance models to public domains but catalyzes the search for new analytical tools. The empirical continuation emerging from this critique will thus move from **diagnosis to design**.

from identifying model failure to constructing an evidence-based, theoretically coherent framework for **value-based financial sustainability** in healthcare.

7. Recommendations for Improvement

The analytical and theoretical limitations identified in *Karakolias (2025)* underscore the urgent need for a more rigorous and context-appropriate framework for studying financial sustainability in public healthcare.

The author's pioneering attempt to apply a classical corporate finance model to a Beveridgean system highlights a methodological paradox: traditional bankruptcy-based tools can expose inconsistencies but cannot explain them.

This chapter proposes a set of **concrete improvements**, organized across four dimensions-conceptual, methodological, governance-related, and multidimensional integration-to guide future research, including the forthcoming empirical paper using Portuguese data.

7.1 Redefinition of Financial Distress

The first and most fundamental improvement concerns the **conceptual redefinition** of what constitutes *financial distress* in non-profit and public organizations. In corporate finance, distress represents a transitional phase toward bankruptcy, characterized by liquidity shortages, leverage escalation, and asset devaluation. In the public sector, however, such trajectories rarely culminate in insolvency; instead, they manifest as **chronic fiscal imbalances** and **stakeholder externalities**. Thus, public-sector financial distress should be redefined not as a binary “failure event” but as a **continuous condition of unsustainable dependency** on external rescue mechanisms.

A more appropriate conceptualization would view distress as a **multi-actor process** with both financial and social dimensions. Instead of bankruptcy risk, the relevant dependent variable becomes the **degree of financial autonomy** or **vulnerability to fiscal intervention**. This could be operationalized through the following conceptual dimensions:

1. **Liquidity Risk** – Persistent delays in supplier payments, cash flow deficits, and overreliance on credit extensions.
2. **Subsidy Dependence** – Proportion of operational revenue derived from central transfers, capturing the extent of fiscal reliance on government injections.
3. **Transferred Distress** – The displacement of financial pressure from the institution to external stakeholders-suppliers (payment delays), employees (wage freezes), and patients (service reductions).

4. **Temporal Sustainability** – The organization’s ability to maintain solvency and service levels over time without additional bailouts.

This redefinition aligns with **Kornai’s (1986) soft budget constraint theory**, reframing financial distress as a function of **institutional dependency and intertemporal inefficiency** rather than impending bankruptcy. It transforms the analytic question from “*Will the hospital fail?*” to “*At what cost and through what mechanisms is its viability maintained?*”

Such a reconceptualization forms the intellectual foundation for the **Public Hospital Financial Sustainability Index (PHFSI)** developed in the subsequent empirical paper—an index explicitly designed to quantify financial sustainability in the absence of market-based failure mechanisms.

7.2 Methodological Enhancements

The methodological redesign of future studies should focus on replacing static, descriptive frameworks with **dynamic, causal, and validated models** capable of capturing temporal evolution and institutional interdependence. Several methodological improvements are essential:

a. Adoption of Panel Data Design

Financial distress evolves over time, influenced by cyclical funding patterns, policy reforms, and managerial turnover.

Using **panel data** (e.g., 2017–2024 for Portuguese hospitals) allows for:

- Estimating **within-hospital dynamics** and identifying persistence of financial imbalance.
- Controlling for unobserved heterogeneity (e.g., size, mission, case-mix).
- Testing lagged effects of governance and policy interventions.

Panel methods—such as **fixed-effects** and **random-effects** models—would replace the static cross-sectional snapshot used by Karakolias (2025) with a temporally sensitive analysis.

b. Implementation of Causal Inference Strategies

To identify the determinants of financial sustainability, causal inference is indispensable.

Methods such as **difference-in-differences (DiD)** and **instrumental variables (IV)** can isolate the impact of reforms (e.g., Portugal's 2024 *ULS integration reform*) from confounding factors.

These approaches enable testing of hypotheses like:

- *Does integration of primary and hospital care improve financial sustainability?*
- *Do hospitals with stronger governance structures exhibit slower accumulation of arrears?*

c. Model Validation and Predictive Testing

Unlike the Z"-score, which lacks real-world validation, future models should incorporate **empirical verification** through observable outcomes.

Possible validation targets include:

- Occurrence of **government interventions** (capital injections, audits, or mergers).
- Sudden surges in **overdue liabilities** or **supplier defaults**.
- Changes in **operational or clinical performance** following financial adjustments.

Predictive accuracy should be tested using **Receiver Operating Characteristic (ROC)** curves or **out-of-sample forecasts**, ensuring that models possess genuine predictive value rather than descriptive adequacy.

d. Data Transparency and Reproducibility

To align with modern research standards, all data sources and transformation steps must be openly documented.

Utilizing **public transparency portals**, publishing **data dictionaries**, and providing **replication code** (in R or Python) will enhance credibility and allow international replication-particularly across European Beveridgean systems.

Through these methodological enhancements, future work can evolve from an *exploratory diagnostic* to a **causally robust explanatory model** of financial sustainability.

7.3 Governance Variable Refinement

A key weakness of Karakolias (2025) lies in the **simplistic treatment of governance** as a categorical attribute. The relationship between governance quality and financial outcomes cannot be captured through binary variables (e.g., CEO gender) or organizational size alone.

To align with **agency theory**, **stewardship theory**, and **institutional economics**, governance must be operationalized as a **multidimensional construct** encompassing both structure and process.

The following refinements are recommended:

1. Board Composition and Expertise

Include measures of financial and clinical expertise among board members.

Quantify the proportion of independent members relative to political appointees.

Capture diversity not only by gender but also by professional background and experience.

2. Audit and Oversight Mechanisms

Frequency of internal audits and the existence of external audit committees.

Implementation of performance monitoring systems tied to financial outcomes.

Transparency indicators (public disclosure, compliance with IPSAS).

3. Managerial Incentives and Tenure

Length of CEO tenure and turnover rate as proxies for stability and accountability.

Presence of performance-based incentives or contracts linked to sustainability metrics.

4. Decision-Making Autonomy

Degree of fiscal autonomy in procurement and capital investment.

Ability to reallocate budget lines or retain surpluses for reinvestment.

Developing a **Governance Quality Index (GQI)** that aggregates these indicators would provide a robust, theory-driven measure to test hypotheses about the governance–sustainability nexus.

For instance, the empirical continuation will test whether a **one-standard-deviation increase in GQI** improves the PHFSI score or reduces dependency on subsidies.

Such refinement elevates governance from a contextual descriptor to a **causal explanatory mechanism** within the analytical model.

7.4 Integration of Multi-Dimensional Sustainability Metrics

One of the most profound conceptual oversights in the Greek study is the **exclusive reliance on financial ratios** as indicators of institutional performance. In public healthcare, financial sustainability cannot be isolated from **operational efficiency, service quality, and equity of access**-dimensions that together define the system's ability to deliver long-term value.

To achieve a more comprehensive understanding, future research should adopt a **multi-dimensional sustainability framework** that integrates both financial and non-financial indicators.

a. Financial Dimension

- Liquidity and solvency ratios adjusted for public-sector accounting (e.g., *cash collections / accrued revenues*).
- Subsidy dependence ratio: share of operating costs covered by transfers.
- True leverage ratio: total liabilities + expected future subsidies / retained earnings.

b. Operational Dimension

- Resource utilization efficiency: *cost per discharge, bed occupancy rates, average length of stay*.
- Productivity indicators: *staff per bed, patient throughput per FTE*.
- Payment discipline: *supplier payment delays, arrears growth rate*.

c. Quality and Service Dimension

- Clinical outcome metrics: *readmission rates, infection rates, mortality standardized ratios*.
- Access and equity: *waiting times, regional disparities, patient satisfaction*.
- Patient-reported outcomes (PROMs/PREMs), where available, to capture perceived service value.

By combining these dimensions, researchers can construct composite indices that reflect both **financial viability** and **mission performance**. The proposed PHFSI model in the Portuguese study will integrate these dimensions to represent a **balanced scorecard of sustainability**-a departure from ratio-centric methods toward holistic value-based evaluation.

This approach aligns with modern frameworks such as **Value-Based Healthcare (Porter & Teisberg, 2006)** and **Triple Bottom Line accounting**, embedding financial analysis within the broader pursuit of public value.

Ultimately, it restores the link between **corporate finance theory** (focused on value creation and risk management) and the **public health mission** (focused on service sustainability and equity).

Synthesis

The four sets of recommendations-conceptual redefinition, methodological redesign, governance refinement, and multidimensional integration-together outline a pathway toward a **new generation of financial analysis in public healthcare systems**. They respond directly to the shortcomings identified in Karakolias (2025) by shifting the analytical paradigm:

Dimension	Greek Study (Current)	Recommended Improvement (Future)
Conceptualization	Bankruptcy-based distress (Altman)	Fiscal-dependency-based sustainability (PHFSI)
Methodology	Cross-sectional, descriptive	Panel data, causal, validated
Governance	Simplistic categorical variables	Multi-indicator Governance Quality Index
Performance Scope	Purely financial	Financial + operational + quality integration

These recommendations will be implemented concretely in the **empirical continuation**, transforming the Greek case's conceptual shortcomings into a **comprehensive, evidence-based model** for understanding financial resilience in Beveridgean systems.

8. Integration with Advanced Corporate Finance and Empirical Continuation

This chapter positions *Karakolias (2025)* within the intellectual architecture of the **Advanced Corporate Finance (CFE)** course, linking the critical review to both the core theoretical constructs of corporate finance and the forthcoming **empirical continuation** focused on Portuguese public hospitals.

The aim is to demonstrate that the analytical exercise extends beyond critique, functioning as an **academic bridge**-transforming a static descriptive study into a dynamic, theory-driven research project that aligns with the course’s emphasis on financial modeling, governance, and sustainable value creation.

8.1 Alignment with CFE Topics

The *Advanced Corporate Finance* syllabus integrates the pillars of **financial theory, quantitative analysis, and corporate governance**, all of which converge in the reviewed article, albeit in unconventional ways. *Karakolias (2025)* provides a platform to interrogate the limits of these concepts when applied to **public-sector finance**. Its relationship to key CFE topics can be mapped as follows:

CFE Topic	Relevant Concept	Article Connection	Pedagogical Insight
Capital Structure Theory	Trade-off between debt and equity; cost of capital	Greek hospitals exhibit near-zero leverage and subsidy-funded “equity,” challenging conventional optimization logic.	Illustrates how tax-exempt, state-backed entities invalidate the trade-off model’s predictions.
Financial Distress and Bankruptcy Models	Altman Z-score; default prediction	Applies Z"-score to non-market hospitals, producing artificially high solvency scores.	Demonstrates boundary conditions of market-based distress models; motivates redefinition of “financial failure.”
Corporate Governance and Agency Theory	Managerial discretion, gender diversity, size, board structure	Explores governance variables (CEO gender, size) but without causal interpretation.	Opens discussion on governance mechanisms in non-shareholder institutions and the

CFE Topic	Relevant Concept	Article Connection	Pedagogical Insight
			multiplicity of principals in public finance.
Empirical Finance Methods	Model specification, causality, identification strategies	Relies solely on non-parametric tests; lacks causal design.	Highlights necessity of econometric rigor (panel data, DiD, IV) in validating financial theories.
Sustainable Finance and ESG Integration	Value creation beyond profitability	Implies need to link financial health with social outcomes in public healthcare.	Bridges ESG principles and value-based healthcare metrics.

Through this mapping, it becomes evident that Karakolias’s work-though methodologically limited-serves as a **didactic counterexample** that reinforces central learning outcomes of the course. It challenges the direct transposition of corporate finance paradigms to non-profit environments and compels the student to rethink value, risk, and performance measurement within **mission-driven institutions**.

8.2 Theoretical Link to Empirical Work

The critical review forms the **theoretical foundation** for the empirical paper that will extend the analysis to Portuguese hospitals under the **Serviço Nacional de Saúde (SNS)**. Specifically, the review’s findings provide three crucial theoretical insights that shape the design of the upcoming research:

1. Boundary Conditions of Corporate Finance Models

The failure of the Z"-score to predict financial fragility underscores that traditional corporate finance theory requires contextual adaptation. The empirical project will translate *financial distress* into *financial sustainability*, consistent with **soft budget constraint** and **agency frameworks**, merging public finance theory with corporate principles.

2. Integration of Governance and Fiscal Dependency Theories

The review’s critique of superficial governance measures motivates a shift toward **institutionally grounded governance indicators**. The forthcoming research will integrate **agency theory** (Jensen & Meckling, 1976) with **stewardship theory**, reflecting the dual nature of public healthcare management-balancing accountability to multiple stakeholders with service-oriented missions.

3. Embedding Value-Based Perspectives

The redefinition of financial distress aligns with **value-based healthcare (VBHC)** principles, linking financial outcomes to social value. The empirical paper will position financial sustainability as a necessary condition for long-term value creation in health systems—thus merging corporate finance’s efficiency logic with public health’s equity mandate.

In essence, this review acts as a **conceptual handoff** between descriptive critique and model-building. It identifies what corporate finance theory overlooks in public institutions, while the empirical continuation seeks to **rebuild the analytical scaffolding** required to explain and predict sustainability in such contexts.

8.3 Methodological Transition

A second bridge between the review and the empirical paper lies in **methodological evolution**. Where *Karakolias (2025)* offered a static, diagnostic exercise, the empirical continuation will employ a **dynamic, causal, and multi-dimensional methodology** designed to overcome each identified weakness.

Dimension	Greek Study Approach	Empirical Continuation (Portuguese Study)	Expected Outcome
Data Design	Single-year cross-section (2022)	Longitudinal panel (2017–2024) using SNS Transparency Portal	Captures trends, persistence, and temporal causality
Dependent Variable	Altman Z"-score (bankruptcy proxy)	Public Hospital Financial Sustainability Index (PHFSI)	Redefines distress as fiscal sustainability
Governance Measurement	Categorical descriptors (gender, size)	Governance Quality Index (board expertise, audit, autonomy)	Theoretically grounded, continuous indicator
Analytical Method	Non-parametric tests	Fixed-effects, dynamic GMM, and DiD models	Causal inference and hypothesis testing
Validation Criterion	None (no bankruptcies)	Predictive validation via intervention events (bailouts, capital injections)	Empirical falsifiability and robustness

Dimension	Greek Study Approach	Empirical Continuation (Portuguese Study)	Expected Outcome
Performance Scope	Financial ratios only	Financial + operational + quality dimensions	Multi-dimensional measurement of sustainability

Through this transition, the empirical paper will operationalize the **Public Hospital Financial Sustainability Index (PHFSI)**, integrating liquidity realization, subsidy dependence, stakeholder pressure, and performance indicators.

This multidimensional framework will permit **causal testing** of governance effects and **policy evaluation**, such as the impact of the 2024 ULS integration reform on financial resilience.

In doing so, the empirical study will transform the diagnostic critique of Karakolias into an actionable, reproducible analytical model consistent with contemporary standards in **empirical corporate finance research**.

8.4 Expected Academic Contribution

The combined contribution of this **critical review** and its **empirical continuation** lies in the creation of a coherent analytical pathway that **extends corporate finance theory** into the public healthcare domain while maintaining methodological integrity.

The project as a whole generates contributions at four levels-conceptual, methodological, empirical, and normative:

1. Conceptual Contribution:

Introduces the notion of “*financial sustainability under soft budget constraints*” as a redefinition of financial distress for public entities.

Repositions financial resilience as the equilibrium between liquidity, autonomy, and value creation-expanding the corporate finance lexicon beyond profit-oriented contexts.

2. Methodological Contribution:

Advances the field by developing and validating the **PHFSI** as a measurable, data-driven alternative to the Z"-score for public institutions.

Demonstrates how empirical finance tools (panel data, DiD, IV) can be repurposed to analyze non-market organizations.

3. Empirical Contribution:

Provides cross-national evidence by comparing Greece and Portugal, illustrating how governance and fiscal context shape sustainability.

Validates the conceptual framework using open, reproducible datasets-enhancing transparency and replicability in public finance research.

4. Normative and Policy Contribution:

Offers policymakers and hospital administrators a new diagnostic toolkit to monitor fiscal dependency, predict intervention needs, and design governance reforms.

Aligns with **EU policy priorities** on sustainable public finance, open data, and health system resilience.

By linking these contributions, the work exemplifies the **CFE learning outcomes**: the ability to apply advanced financial theory to complex, real-world systems; to critique and refine existing models; and to use empirical methods to generate actionable insights.

The transition from *Karakolias (2025)* to the forthcoming Portuguese study thus mirrors the pedagogical progression from **theoretical critique** to **empirical innovation**, embodying the research-oriented ethos of advanced corporate finance education.

9. Conclusion

9.1 Summary of Evaluation

The critical review of *Karakolias (2025)*, “*Rethinking Financial Distress in Public Healthcare: Evidence from Greek Hospitals’ Governance Attributes*,” reveals a study that is **ambitious in scope yet limited in theoretical depth and methodological precision**. Its principal contribution lies in exposing a paradox: Greek public hospitals appear financially sound according to classical corporate finance metrics, despite clear signs of structural fragility, subsidy dependence, and liquidity shortages. This paradox serves as a valuable empirical and conceptual warning—that tools designed for profit-maximizing firms cannot be uncritically transposed to non-market, mission-driven organizations.

From a **theoretical standpoint**, the paper makes a modest but important contribution by identifying the boundary conditions of financial distress models when applied outside their natural environment. It highlights how the Altman Z"-score’s predictive logic breaks down in the absence of bankruptcy risk, implicitly calling for a broader conception of fiscal distress rooted in institutional economics and public finance theory.

From a **methodological perspective**, however, the study’s cross-sectional design, limited operationalization of governance, and lack of causal or validation tests undermine its explanatory potential. The use of a single-year dataset (2022) prevents dynamic interpretation, while the absence of outcome validation (such as state bailouts or supplier arrears) restricts its empirical robustness. Its reliance on non-parametric tests and descriptive ratios yields insights into patterns but not mechanisms.

Nevertheless, the paper’s **relevance remains undeniable**. It opens a necessary dialogue between corporate finance and public-sector management at a time when healthcare systems worldwide face mounting fiscal stress. By testing the limits of existing models, *Karakolias (2025)* provides the conceptual catalyst for developing new analytical frameworks that capture the economic realities of public healthcare providers-entities that cannot fail but can nevertheless experience systemic distress.

9.2 Bridge to Empirical Study

The shortcomings identified in this critical review provide a **clear research roadmap** for the empirical continuation within the Advanced Corporate Finance course. Where *Karakolias (2025)* ends-with a call to “redefine financial distress” in public entities-the forthcoming study begins, translating this conceptual imperative into a **dynamic, empirically testable model**.

The next research stage will focus on **Portuguese public hospitals**, operating within another Beveridgean health system, to test the hypotheses derived from the Greek case.

Using longitudinal data (2017–2024) from the **SNS Transparency Portal**, the empirical paper will address each methodological limitation observed in the Greek study by:

1. **Adopting a Dynamic Framework:** Introducing **panel data econometrics** to capture temporal evolution, persistence, and causality in financial distress processes.
2. **Redefining the Dependent Variable:** Constructing the **Public Hospital Financial Sustainability Index (PHFSI)**—a multi-dimensional measure encompassing liquidity realization, subsidy dependence, stakeholder pressure, and performance outcomes.
3. **Grounding Governance Measurement:** Developing a **Governance Quality Index (GQI)** aligned with agency and stewardship theory, incorporating board expertise, audit frequency, managerial tenure, and transparency.
4. **Implementing Causal Identification:** Employing **difference-in-differences (DiD)** and **instrumental variable (IV)** approaches to isolate the effects of governance reforms and policy changes, such as the 2024 *ULS integration reform* in Portugal.
5. **Validating Model Predictions:** Testing whether PHFSI scores predict **actual fiscal interventions** (bailouts, capital injections, or arrears clearance), thus providing falsifiable evidence of model accuracy.

Through these advancements, the empirical paper will transform *Karakolias's* diagnostic exercise into a **comprehensive, causal, and policy-relevant framework** for measuring and managing financial sustainability in public healthcare systems.

The transition from Greece to Portugal represents not merely a change in dataset but an **evolution in analytical philosophy**—from a static assessment of accounting ratios to a dynamic, theory-informed model of institutional resilience.

9.3 Final Reflection

The evaluation of *Karakolias (2025)* reinforces a broader insight central to both corporate finance and public policy: **financial theory must evolve when applied beyond its market origins**. Public hospitals, universities, and other mission-driven institutions operate within ecosystems governed by political priorities, ethical mandates, and non-market dependencies. In such environments, traditional concepts-leverage optimization, distress prediction, shareholder value-require reinterpretation through the lens of **stakeholder value and long-term sustainability**.

Adapting corporate finance models to these contexts is not a dilution of the discipline but an **expansion of its relevance**. By integrating principles of **soft budget constraints, governance accountability, and data-driven performance measurement**, corporate finance can contribute to the design of institutions that are both financially responsible and socially purposeful.

This review, therefore, performs more than an academic critique-it represents a methodological and philosophical inflection point. It identifies how financial models can misrepresent value in public systems and charts a path toward a new generation of analytical tools capable of reconciling fiscal prudence with social mission.

In conclusion, *Karakolias (2025)* should be read not as a failed experiment but as a **necessary provocation**-an invitation to rethink the very meaning of financial health when profit is not the objective and bankruptcy is not the threat. The forthcoming empirical study will accept that invitation, operationalizing this rethinking through rigorous, transparent, and replicable research. Together, the two works-one diagnostic, the other reconstructive-will contribute to a more inclusive and context-aware **theory of corporate finance for public-sector sustainability**.

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Annex A

The following table summarizes the main academic contributions that contextualize *Karakolias (2025)* within the intersection of **corporate finance**, **public-sector economics**, and **healthcare financial management**.

Each entry identifies the **author(s)**, **publication year**, **research goal**, **data and scope**, **methodology**, and **key conclusions**. Together, these studies provide a comparative framework that highlights both the originality and the limitations of the Greek analysis and informs the conceptual and empirical design of the forthcoming Portuguese study.

Author(s) & Year	Goal / Research Question	Data and Scope	Methodology / Model Used	Main Conclusions and Relevance
(Altman, 1968; Altman & Hotchkiss, 2010; Casey et al., 1984)	Develop and refine a quantitative model to predict bankruptcy risk in private firms.	U.S. manufacturing firms, 1946–1965 (various samples).	Discriminant analysis; Z-score model combining five financial ratios.	Z-score effectively predicts corporate failure in competitive markets; assumes market discipline and profit maximization. Foundation for later misapplications in public sector studies.
Whitaker (1999)	Identify early warning indicators of financial distress in corporate settings.	U.S. firms (1980–1990s), longitudinal data.	Panel regression and time-to-failure analysis.	Financial distress is a dynamic process; early signals appear 3–5 years before bankruptcy. Demonstrates necessity of time-series data.

Author(s) & Year	Goal / Research Question	Data and Scope	Methodology / Model Used	Main Conclusions and Relevance
Kornai (1986)	Conceptualize “soft budget constraints” in socialist and post-socialist economies.	Theoretical and empirical analysis of state enterprises.	Institutional economics framework; conceptual modeling.	Public entities anticipate bailouts, undermining financial discipline. Relevant to interpreting public hospital dependency on subsidies.
Maskin & Xu (2001)	Formalize soft budget constraint theory under market transition conditions.	Transition economies (China, Eastern Europe).	Game-theoretic modeling.	Confirms that bailout expectations generate inefficiency. Provides theoretical underpinning for redefining financial distress in state-funded organizations.
Ramamonjiarivelo et al. (2015)	Examine whether privatization mitigates financial distress in U.S. public hospitals.	422 U.S. public hospitals (2000–2009).	Logistic regression; financial ratio analysis.	Hospitals in financial distress are more likely to privatize; ownership structure influences sustainability. Shows governance and market exposure as determinants of recovery.
Dubas-Jakóbczyk et al. (2020)	Assess financial performance and debt structure of Polish public hospitals.	247 Polish hospitals, 2010–2017.	Ratio analysis; trend evaluation.	Chronic deficits and overdue liabilities reflect systemic underfunding. Highlights European parallels to Greek case.
Matos, Ferreira, & Pedro (2021)	Analyze efficiency, quality, and sustainability of Portuguese public hospitals.	74 hospitals (2015–2018) using SNS financial and operational data.	Data envelopment analysis (DEA) and financial ratio analysis.	Efficiency and quality partially decoupled; many hospitals financially fragile despite good operational performance. Shows

Author(s) & Year	Goal / Research Question	Data and Scope	Methodology / Model Used	Main Conclusions and Relevance
Karakolias (2025)	Test applicability of Altman's Z"-score to Greek public hospitals; examine governance attributes.	90 Greek public hospitals (2022).	Z"-score calculation; non-parametric tests (Mann–Whitney, Kruskal–Wallis).	need for multidimensional assessment. All hospitals appear solvent (mean Z = 15.7), yet are subsidy-dependent. Concludes that financial distress must be redefined for public entities. Basis for critique in current review.
OECD (2023)	Monitor health financing trends and system efficiency across Europe.	OECD Health Accounts 2010–2021.	Comparative descriptive analysis.	Increasing hospital debt and delayed payments across EU. Confirms that Greece and Portugal share structural fiscal challenges.
Porter & Teisberg (2006)	Introduce the value-based healthcare (VBHC) model.	Theoretical, with case evidence from U.S. healthcare systems.	Conceptual framework emphasizing outcomes per cost unit.	Aligns performance measurement with patient value rather than financial inputs. Provides normative foundation for integrating quality into fiscal metrics.
Elkington (1997)	Propose “Triple Bottom Line” accounting for sustainable organizations.	Multi-sector conceptual work.	Theoretical development of sustainability framework.	Expands financial analysis to include social and environmental dimensions; applicable to public hospitals pursuing long-term value creation.

Interpretive Notes

1. **Chronological and Thematic Continuity:**

The literature evolves from the foundational work on **corporate bankruptcy** (Altman, 1968) toward more recent debates on **public-sector sustainability** and **governance reform**. *Karakolias (2025)* occupies a transitional position-testing corporate models in the public sphere but not yet integrating the broader sustainability paradigm.

2. **Comparative European Context:**

Studies from **Portugal (Matos et al., 2021)** and **Poland (Dubas-Jakóbczyk et al., 2020)** collectively show similar fiscal weaknesses-soft budget constraints, delayed payments, and governance inefficiencies-suggesting that the Greek case is not exceptional but systemic to Beveridgean systems.

3. **Governance as an Emerging Determinant:**

The gender and managerial findings of *Karakolias (2025)* signal the growing recognition of governance diversity and institutional design as key variables in explaining sustainability outcomes.

4. **Shift Toward Multidimensional Models:**

Contemporary research (Porter & Teisberg, 2006) advocates for frameworks that combine financial and non-financial metrics. The forthcoming Portuguese empirical study responds to this call by operationalizing such integration through the **Public Hospital Financial Sustainability Index (PHFSI)**.

Synthesis

The comparative literature demonstrates that while **financial ratio analysis** remains useful for benchmarking, it is insufficient for understanding sustainability in tax-funded systems. Across countries, public hospitals exhibit a recurring pattern: **accounting solvency masking fiscal fragility**.

The next step, embodied in the empirical continuation of this review, is to move beyond static models toward **dynamic, multi-dimensional, and causally grounded analyses** that bridge **corporate finance theory** and **public-sector value creation**.

